

Context-Aware Recommendation of Learning Resources Using Rules Engine

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Abstract – Although numerous studies have been conducted on learning resources recommendation in E-Learning, research extending this investigation into usage of users' contextual information is rare. This paper presents an innovative architecture of an intelligent personalized context-aware recommendation system in an E-Learning environment. The system offers users by recommending learning materials, tutors, or other learners with common interests combining users' social tags from external social network and tags about learning materials from autologous collaborative tagging system. A Rules Engine is used to manage a set of rules for each user to achieve personalized recommendation. The rules which define the relationship between tags construct the relationship graph model for users and learning resources thus graph-based collaborative filtering recommendation methods can be implemented. The rules will be adjusted depending on the user's feedbacks of previous recommendations.

Keywords – *Personalized and Context-Aware Technology-Enhanced Learning; Collaborative and Social Tagging; Intelligent Recommendation; Rules Engine; Social network;*

I. INTRODUCTION

In recent conceptual studies, the role of recommendation systems to enhance the social and collaboration aspects of E-Learning systems has been outlined [1], and a number of recommendation systems have been introduced in order to propose learning resources to users [2]. Considering the tremendous learning resources online, such systems could potentially play an important educational role and help people build new relationships and enable them to join communities based on their personalized preferences.

With the enormous development of E-Learning, the concept of 'context-aware ubiquitous learning' has been proposed to facilitate a seamless ubiquitous learning environment that supports learning without constraints of time or place [3]. However, majority of existing recommendation systems focus on recommending the most relevant items to individual users and do not take into consideration any contextual information. In recent years, several efforts have been made to use contextual information in recommendation systems. These researches can be broadly categorized into two types depending on different approaches to use contextual information. The first approach directly uses user's contextual information (e.g., current mood or location) to search a certain space of resources and present the best matching resources to the user. This

approach has been widely used by recommendation systems based on mobile devices [7]. The other approach which represents a more recent trend attempts to model and learn users' context-sensitive preferences. It typically either adopts existing recommendation methods or applies various intelligent techniques from data mining and machine learning. Many of these studies utilize social tags and semantics of resource metadata to enhance recommendation systems [4][6]. On the other hand, social networks like Facebook are rich with detailed contextual information describing an individual's preferences and relationships. We believe that this contextual information can make recommendation more feasible and effective.

In this paper, we propose an innovative architecture of an intelligent context-aware recommendation system which provides personalized recommendation of Learning Resources. Our study mainly contributes in two aspects. The first aspect is the novel way of extracting users' social tags from external social networks for recommending learning resources to users. Prior research has addressed parts of our vision [5], but as far as we are aware, no single literature has used users' social tags obtaining from external social network as contextual information in recommendation system. Another contribution is outlining a Rules Engine which manages a set of rules respectively for each user to offer personalized recommendation. The rules define the relationships between tags. For each user, his rule set is unique as it will be adjusted according to the user's feedbacks. The rule set is used to construct a heterogeneous object network model for users, learning materials and tags, and then the general Random Walk based recommendation methods [8] can be applied based on the model.

II. RECOMMENDATION SCENARIO

As shown in Figure 1, users act as learner or tutor. A tutor who is expert at a specific domain may also be a learner in other domains. Both of learners, tutors, and learning materials have its semantic tags. The learning materials' tags are generated by the collaborative effort of users in collaborative tagging system, a part of user social tags is also obtained from the system but such tags are often not abundant enough to effectively describe the semantic meaning of users' preferences and related terms. Therefore we mainly extract user social tags from external social networks like Facebook, Twitter, and LinkedIn etc. while the collaborative tagging system is used as a supplement.

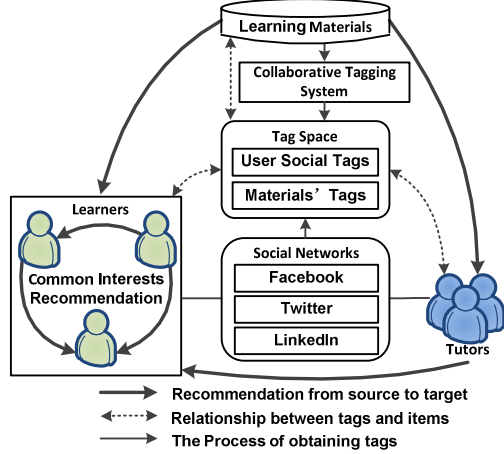


Figure 1. The scenario of learning resources recommendations.

The user social tags contain users' statistical information, topics of interests, and role tags. Users' statistical information mainly includes users' basic properties such as age, education and location etc. Tags about topics of interests describe users' domain interests and learning purposes. The role tags are special for identifying whether the user is a tutor in a specific domain. Meanwhile the tags of learning materials can be categorized into two types. The most basic type is tags about its subjects, domains, and publication date etc. The other one includes the evaluation and impression of the learning material from collaborative tagging. Learning materials, learners, and tutors are associated by tags.

Our recommendation system offers three types of recommendations. The first one is to recommend learning materials to learners and tutors. The second one is to recommend tutors to learners for the learners' need of strengthening their abilities by interactive communication with domain experts. The last one is to recommend users to users with common interests, it helps to provide a collaborative environment and promote users to join communities based on interests.

III. RULES ENGINE & ALGORITHM

In our system, the recommendation involves significant conditional branching or decision-making when serve different users. Meanwhile, the recommendation rules for each user should change gradually according to feedbacks. Therefore, we will gain a significant advantage from rules engine for offering personalized recommendation and avoiding the complexity of maintaining the business logics. Then, user's rules are used to establish the relationship model and the algorithm is applied.

A. Definition of a Rule

A rule is defined as the relationship degree between a pair of tags. Suppose T_a , T_b are tags of learner A and learning material B respectively. T_a is related to T_b by the rule R which is described as weight (T_a, T_b) ; the weight is a real number in a finite interval. Thus, the rule R establishes a relationship between learner A and learning material B.

B. Relationship Modeling

Based on the user's rule set, we can gain a full view of the relationship network of entire items (learners, tutors, learning materials). We model it as a heterogeneous object network shown in Figure 2.

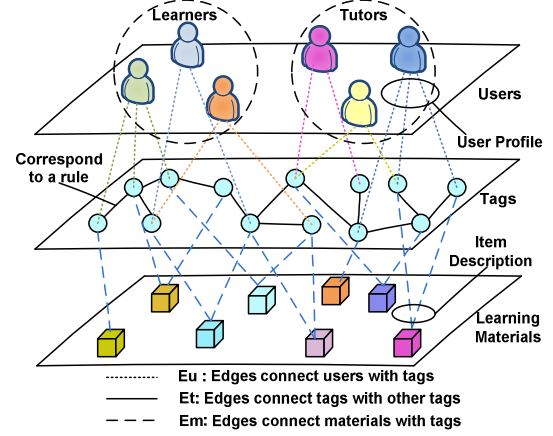


Figure 2. The heterogeneous object network model.

In Figure 2, users are connected with their social tags which describe user's profile, interests, social relationships and behaviors. The connections among users are reflected as the associability of the two users' social tag set. Similarly, learning materials connect to their tags representing their unique features. The connections between users and learning materials are also the associability of their tag set. We could describe the whole networks as $G = \{V (V_u, V_m, V_t), E (E_u, E_m, E_t)\}$; V is the set of nodes including V_u (users), V_m (learning materials) and V_t (tags); E is the set of edges including E_u (edges connect users with tags), E_m (edges connect materials with tags) and E_t (edges connect tags with other tags). Edges' weight in E_u or E_m is defined by tag's preference for the owner. While an edge in E_t corresponds to a rule in rules engine.

C. Recommendation Algorithm

Once the heterogeneous object network is established, graph-based algorithms can be applied to generate more precise recommendations by simultaneously considering the information of object properties and the relevancies between them. A Random Walk with Restart (RWR) algorithm can be deployed in the heterogeneous object network model and the recommendations are generated according to the similarity and homophily of the objects [8].

Because our model contains all the users and learning materials as nodes, for any two nodes, we can obtain the stationary probability of random walking (PR) from one to another. Thus, all types of recommendations we mentioned above can be achieved based on the ranking of PR.

D. Intelligent Learning Based on Users' Feedbacks

Users' feedbacks are very useful for re-training the model because of its time effectiveness. When presenting the

recommendation list to users, we allow users to provide feedbacks for recommendations and then use an efficient algorithm to refine user's rule set by leveraging the users' feedbacks. For instance, suppose u is a user with his tag set $T_u : \{tu_1, tu_2 \dots\}$ and m is a learning material with its tag set $T_m : \{tm_1, tm_2 \dots\}$. When we recommend m to u and receive a feedback for m from u , each rule which corresponds to an element of the Cartesian product: $T_u \times T_m = \{ \langle tu, tm \rangle \mid tu \in T_u, tm \in T_m, tu \neq tm \}$ will adjust its weight depending on the feedback is positive or negative.

IV. SYSTEM ARCHITECTURE

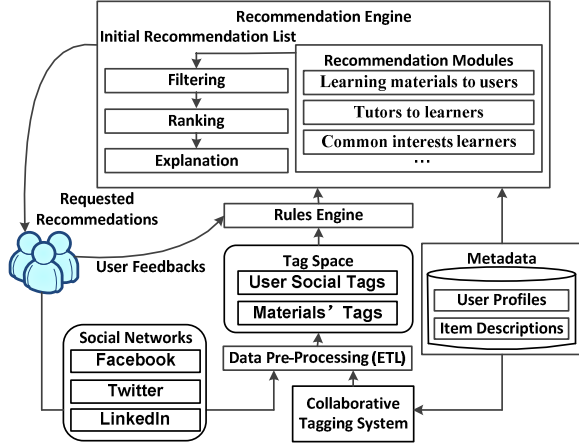


Figure 3. Design of Recommendation System.

As shown in Figure 3, the system consists of three key components.

A. Data Acquisition and Pre-Processing

The system acquires semantic tags from external social networks and autologous collaborative tagging system. The original data is usually full of noisy data and cannot be used directly. A data pre-processing module is implemented to filter the noisy data and establish the tag space which includes user social tags and learning materials' tags.

B. Rules Engine

The rules engine manages a set of rules for each user respectively. In the recommendation process, the rules engine use user's rule set to establish the tags' relationships which are used by recommendation engine to construct the heterogeneous object network with metadata including user profiles and item descriptions. The rules engine will automatically learn from user's feedbacks and adjust related rules to make the recommendations more personalized and more suitable for user's interests and learning goals.

C. Recommendation Engine

The recommendation engine has several modules for different types of recommendations. The recommendation engine uses the input data (relationships among tags, and metadata of users and items) to construct the heterogeneous

object network and execute the recommendation algorithm which has been introduced in Section III to generate the initial recommendation list. Based on the initial list, we obtain final results through Filtering, Ranking, and Explanation modules. The Filtering module filters products which do not fit demands. The Ranking module uses user's feedbacks to improve effectiveness of recommendation. Novelty, content diversity, and time diversity might also be in consideration as ranking facts. The Explanation module gives reasons why we recommend these items.

V. CONCLUSION & FUTURE WORK

In this paper, we outline an innovative architecture for a context-aware recommendation system in an E-Learning environment which can automatically adapt to the current interests, learning goals, and knowledge levels of learners. We introduce user contextual information from external social networks to overcome the lacking of user contextual information and propose a heterogeneous object network model to describe the relationships between users, tags, and items. We use a rules engine to manage a set of rules for each user to offer personalized recommendation. At last, we use users' feedbacks to train the user's rule set making it more intelligent. Further work will focus on how to integrate mobile devices to the system and take use of more contextual information like user's current location.

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